MILITARY SPECIFICATION SHEET

ELECTRON TUBE, NEGATIVE GRID

TYPE 7784

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the electron tube described herein shall consist of this document and the latest issue of Specification MIL-E-1.

DESCRIPTION: Planar triode

See figure 1

Mounting position: Any

Weight: 0.16 ounce (4.725 grams) nominal

ABSOLUTE RATINGS:

Parameter:	Fl	Ef	Eb	Ec	Ehk
Unit:	MHz	V	Vdc	Vdc	Vdc
Maximum:	3,000	6.6	200	0, -10	±50
Minimum:	---	6.0	---	---	---
TEST CONDITIONS:	---	6.3	175	Adjust	---

ABSOLUTE RATINGS:

Parameter:	Ib	Pp	tk	Cooling	T(seal)
Unit:	mAdc	W	sec	---	°C
Maximum:	12	2.0	---	Conduction	150
Minimum:	--	---	---	---	---
TEST CONDITIONS:	10	---	180	Conduction	---

GENERAL:

Qualification - Required

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METHOD	REQUIREMENT OR TEST	CONDITION	AQL (PERCENT DEFECTIVE)	INSPECTION LEVEL OR CODE	SYMBOL	LIMITS		UNIT
						Min	Max	
	Quality conformance inspection, part 1							
1211	Insulation of electrodes	Ebb = -500 Vdc; Ekk = +45 Vdc See figure 2	0.65 0.65	II II	Rgp Rgk	5.0 0.25	--- ---	Meg Meg

1231	Pulsing emission	Ef = 5.5 V;
		Eb = 0;
		Ec = -18 Vdc;
		eb = 390 v (peak-to-peak);
		ec = 46 v (peak-to-peak);
		Rg = 470 ohms;
		Rk = 10 ohms
		See note 1
			0.65	II	Is	3.5	---	mAdc

1261	Electrode voltage (anode)	Ec = 0; Eb/Ib = 10 mAdc	0.65	II	Eb	75	175	Vdc
1301	Heater current		0.65	II	If	260	305	mA
1306	Transconductance		0.65	II	Sm	11,500	---	mhos
4253	Power gain (1)	Ec = 0; Eb/Ib = 10 mAdc; F = 450 ±5 MHz; bandwidth = 9 MHz minimum See note 2	0.65	II	Gain	15.0	---	dB
4260	Noise figure (1)	Ec = 0; Eb/Ib = 10 mAdc; F = 450 ±5 MHz See note 2	0.65	II	NF	---	5.0	dB
	Quality conformance inspection, part 2							
4253	Power gain (2)	Ec = 0; Eb/Ib = 10 mAdc; F = 3,000 ±5 MHz See note 2	---	---	Gain	10.0	---	dB
4260	Noise figure (2)	Ec = 0; Eb/Ib = 10 mAdc; F = 3,000 ±5 MHz See note 2	---	---	NF	---	13.5	dB
1316	Amplification factor		---	---	Mu	85	140	---
1331	Direct-interelectrode capacitance	No voltages; Dwg. 270-JAN	--- --- ---	--- --- ---	Cin Cgp Cout	3.0 1.5 ---	5.0 2.0 0.025	pf pf pf
1336	Heater-cathode leakage	Ehk = +50 Vdc Ehk = -50 Vdc	--- ---	--- ---	Ihk Ihk	--- ---	20 20	Adc Adc
2126	Envelope strain	See note 3	---	---	---	---	---	---

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METHOD	REQUIREMENT OR TEST	CONDITIONS	AQL (PERCENT DEFECTIVE)	INSPECTION LEVEL OR CODE	SYMBOL	LIMITS		UNIT
						Min	Max	
	Quality conformance inspection, part 3							
----	Life-test provisions	Group B Eb = 200 Vdc; Rk/Ib = 10 mAdc; t = 1,000 hours	---	---	---	---	---	---
	Life-test end points:							
1306	Transconductance		---	---	$\frac{Sm}{t}$	---	25	% decrease
4260	Noise figure (1)		---	---	$\frac{NF}{t}$	---	1.0	dB increase
1031	Low frequency vibration	Ec = 0; Eb/Ib = 10 mAdc; Rp = 10,000 ohms; Dwg. 273-JAN See note 4	---	---	Ep	---	100	mAdc
1041	Shock	No voltages; G = 300; Dwg. 274-JAN See note 5	---	---	---	---	---	---
	Post-shock test end points:							
1306	Transconductance		---	---	Sm	11,500	---	mhos
1211	Insulation of electrodes		---	---	Rgp	5.0	---	Meg
1211	Insulation of electrodes		---	---	Rgk	0.25	---	Meg

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NOTES:

1. The 60-cycle voltages eb and ec must be in phase, and the simusoidal waveform should not be distorted. Read average cathode current.

2. An approved noise source shall be used for noise figure measurements. Power input level shall be about -75 dbm for power gain (1), noise figure (1), power gain (2), and noise figure (2).

3. Envelope strain procedure. Tubes shall be tested as specified in Method 2126, except that they shall first be immersed in water at not more than 5°C for 15 seconds, and immediately thereafter subjected to the standard temperature cycle specified in Method 2126. The criteria for air leaks shall be heater current of 315 mA or more.

4. Vibrate in plane perpendicular to the plane of the grid. Test 6 tubes selected at random from the first production lot of each calendar year, and approximately every 90 days of production during the year. If more than one tube fails to pass the specified end points, the test shall become a part of quality conformance inspection, part 2, AQL of 6.5, inspection level S3 on all lots in process. After three consecutive successful submissions, the test shall revert back to the 6-tube quarterly test.

5. Apply the force perpendicular to the plane of the grid from the cathode end. Test 6 tubes selected at random from the first production lot of each calendar year. If more than one tube fails to pass the specified end points, the test shall become a part of quality conformance inspection, part 2, AQL of 6.5, inspection level S3 on all lots in process. After three consecutive submissions, the test shall revert back to a 6-tube annual test.

Custodians

Army - EL
Navy - EC
Air Force - 85

Preparing activity:

Air Force - 85

Review activities:

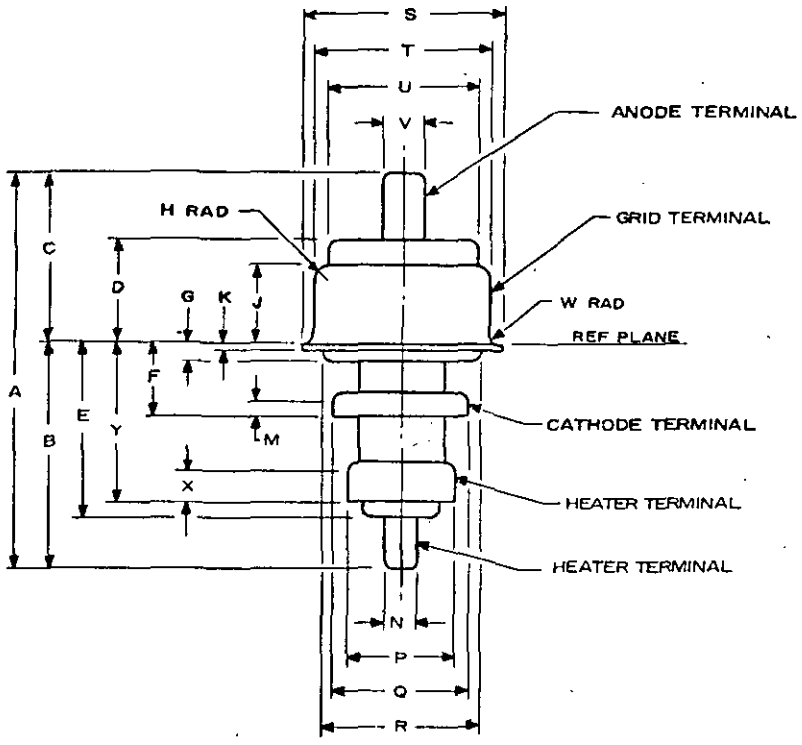
Navy - SH
Air Force - 17, 80

(Project 5960-2800)

User Activities:

Navy - AS, OS, MC, CG
Air Force - 11, 14, 19

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DIMENSIONS

LTR	INCHES		MILLIMETERS		NOTES
	MIN	MAX	MIN	MAX	
QUALITY CONFORMANCE INSPECT, PART 2					a
A	.960	1.040	24.38	26.42	
B	.530	.590	13.46	14.99	
C	.410	.470	10.41	11.94	
D		.272		6.91	
E		.475		12.07	
F	.163	.193	4.14	4.90	
Y	.355	.395	9.02	10.05	
QUALITY CONFORM. INSP., PART 3					b
G		.060		1.52	
H		.030		.76	
J	.190	.210	4.83	5.33	

LTR	INCHES		MILLIMETERS		NOTES
	MIN	MAX	MIN	MAX	
K	.009	.015	.23	.38	
M	.040	.070	1.02	1.78	
N	.059	.065	1.50	1.65	
P	.247	.257	6.27	6.53	
Q	.326	.334	8.28	8.48	
R		.385		9.78	
S	.483	.497	12.27	12.62	
T	.435	.445	11.05	11.30	e
U		.385		9.78	
V	.088	.094	2.24	2.39	f
W		.008		.20	
X	.070	.100	1.78	2.54	

NOTES:

a. THESE DIMENSIONS SHALL BE CHECKED ON A LOT-TO-LOT BASIS UNDER QCI, PART 2 TO AN AQL OF 6.5, USING INSPECTION LEVEL S3, THE AQL IS APPLIED TO EACH DIMENSION SEPARATELY AND IS NOT COMBINED.

b. THESE DIMENSIONS SHALL BE INSPECTED AT THE START OF PRODUCTION AND PERIODICALLY EVERY 3 MONTHS DURING PRODUCTION. THE SAMPLE SIZE SHALL BE SIX TUBES AND THE ACCEPTANCE NUMBER SHALL BE ZERO DEFECTS FOR EACH DIMENSION. FAILURE OF ANY DIMENSION TO MEET QCI, PART 3 REQUIREMENTS SHALL CAUSE THE NON-CONFORMING DIMENSION TO BE INSPECTED ON A LOT-TO-LOT BASIS TO A 6.5 AQL, INSPECTION LEVEL S3. AFTER THREE CONSECUTIVE SUCCESSFUL SUBMISSIONS, THE INSPECTION SHALL REVERT TO 3 MONTH INTERVALS.

c. ALL TUBES ARE TO BE 100% TESTED FOR CONCENTRICITY IN GAGE (SEE FIGURE 3). TUBE SHALL BE ROTATED 360° IN GAGE WITHOUT BINDING.

d. GRID SHELL SEAL SHALL BE COMPLETELY FILLED TO CATHODE CERAMIC SHELF.

e. 'T' DIMENSION SHALL BE MEASURED OTHER THAN ACROSS TUBE DESIGNATION ETCH, AND APPROXIMATELY .100 (2.54 mm) FROM THE REFERENCE PLANE.

f. 'V' DIMENSION SHALL BE MEASURED APPROXIMATELY .100 (2.54 mm) FROM THE END OF THE ANODE TERMINAL.

FIGURE 1. OUTLINE DRAWING OF ELECTRON TUBE TYPE 7784

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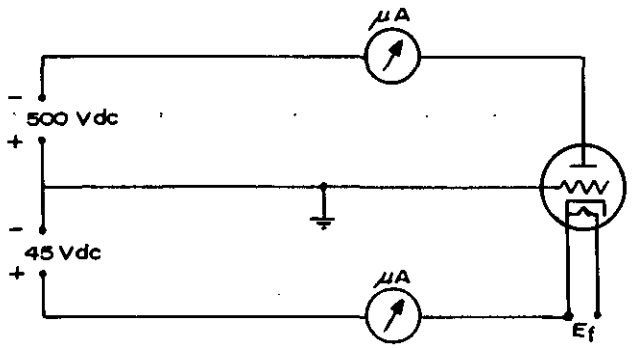
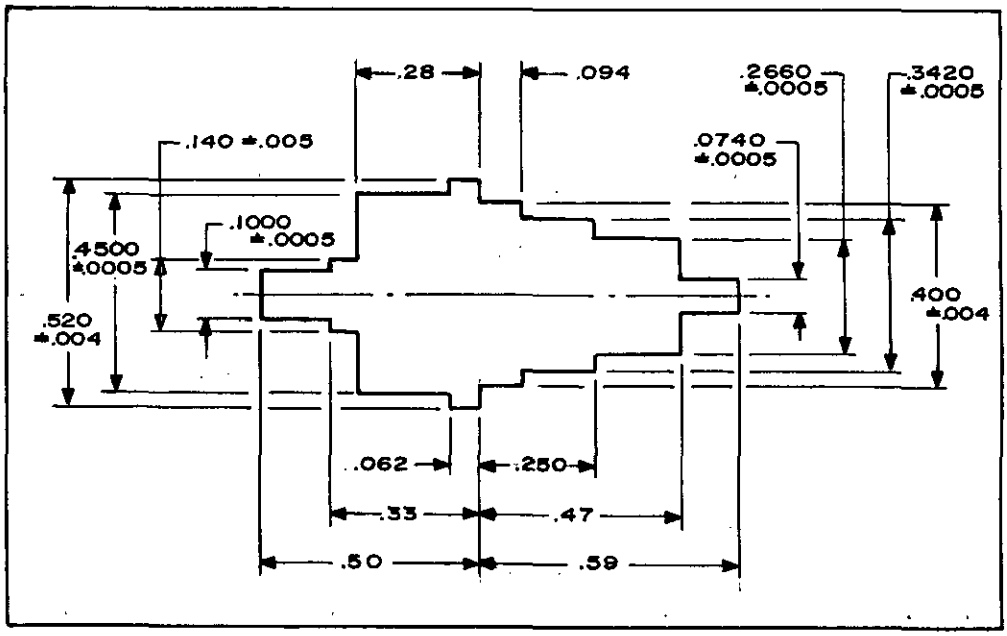


FIGURE 2. Insulation of electrodes test



NOTE:
a. UNLESS OTHERWISE SPECIFIED, TOLERANCE FOR DIMENSION .250 OR LESS ±.008 OVER .25 ±.02.

FIGURE 3 Gage

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